

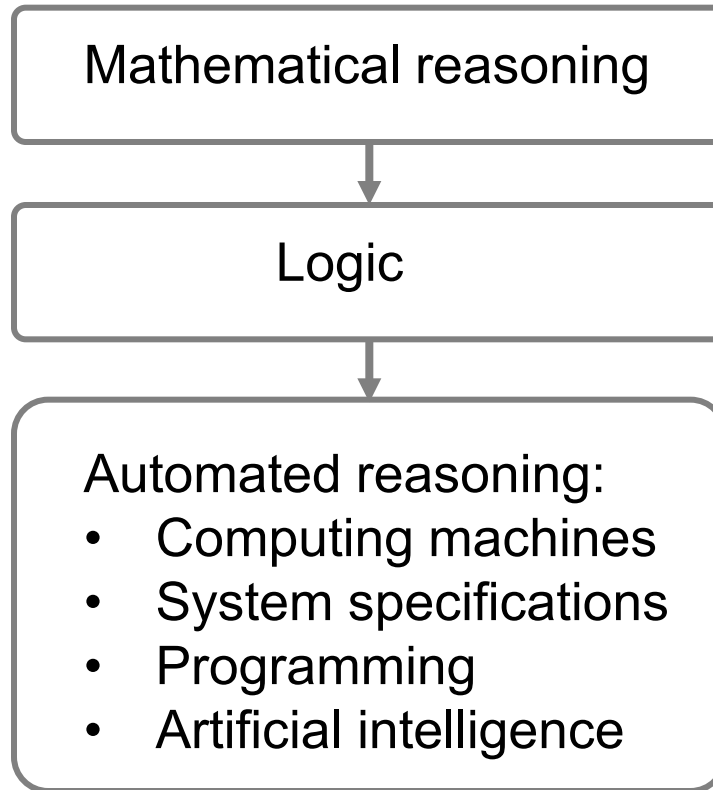
Beregningsteknik for Computer Ingeniører

MM 1: Logic

Topics:

- Propositions.
- Logical operators.
- System specifications
- Propositional calculus

What is logic?



- Correct/ valid mathematical argumentation = proof
- Rules of logic → precise meaning to mathematical statements

Proposition

Definition. Proposition is a declarative statement – something that declares a fact - which can be either true or false, but not both.

Examples – is it a proposition or not?

1. Moon rotates around Earth
2. $2+2=22$
3. $1+2=3$
4. Remember to do your homework
5. Is it raining outside?
6. Computer 1 runs Linux operating system and computer 2 has 1Gb of RAM
7. $x-3=3$

*Aristotle –
Systematic treatises*

*Ordbog
Proposition (engelsk) = Udsagn (dansk)*

Compound propositions

- Many mathematical statements are constructed by combining one or more propositions. Compound propositions are formed from existing propositions using **logical operators**.

Negation

- Definition. Negation $\neg p$ is a statement "It is not the case that p "

*Boyle –
Mathematical
Analysis of Logic*

p	$\neg p$
T	F
F	T

Ordbog

Negation (engelsk) = Negation (dansk)

Truth table (engelsk) = Sanhedstabel (dansk)

Conjunction and disjunction

- Definition. Conjunction $p \wedge q$ of p and q is the proposition "p and q". It is true only when p and q are true.
- Disjunction $p \vee q$ is the proposition "p or q". It is false only when p and q are false.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Ordbog

Conjunction (engelsk) = Konjunktion (dansk)

Disjunction (engelsk) = Disjunktion (dansk)

Exclusive Disjunction

- Definition. The exclusive or $p \oplus q$ is the proposition that is true when exactly one of p and q is true.

The inclusive (disjunction):

“Students who have taken calculus or computer science can take this class.”

The exclusive:

“Students who have taken calculus or computer science, but not both, can enroll in this class.”

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication or conditional statement

- Definition. Implication $p \rightarrow q$ is a statement "If p , then q ".

- If p , then q
- p is sufficient for q
- p only if q
- q when p

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p is the *hypothesis* and q the *consequence (conclusion)* –
asserts q is true on the condition that p holds, hence conditional statement

Ordbog

*Implication (engelsk) = Implikation (dansk) eller
Conditional statement = Betinget udsagn*

Math versus "language"

- Note the difference to natural language. Natural language is often not concise and has cause-and-effect relationships embedded
 - A conditional statement is independent of cause-and-effect relationship (causality) between hypothesis and conclusion
- Example ("if p, then q"):
 - "**if** it is sunny, **then** we will go to the beach" ... is mathematically true in all cases, unless it is sunny (p is true) and we do not go to the beach (q is false), cf. truth table for implication
- In natural language there is a relationship between the hypothesis and the consequence/conclusion, e.g. if p is false (raining) we will not go (q is false) – which seems a consequential truth about the statement.
- The mathematical formulation is more general!

Math versus "language"

- Another example ("q unless $\neg p$ ") :
 - "Maria will find a good job (q) unless she does NOT learn discrete mathematics (p)"
- The sentence implies that Maria needs to learn discrete math (like you), but mathematically the statement is true except when p is true and q is false, so for instance it could also be that Maria did not learn discrete math ($\neg p$ is true) and Maria found a good job.

Systems specifications

- System specifications should be consistent with no conflicting requirements and contradictions, otherwise the system cannot be built:
 - "The diagnostic message is stored in the buffer or it is retransmitted"
 - The diagnostic message is not stored in the buffer"
 - "If the diagnostic message is stored in the buffer, then it is retransmitted"
- Given p: "The diagnostic message is stored in the buffer" and q: "The diagnostic message is retransmitted", it becomes:
 - 1) $p \vee q$; 2) $\neg p$; 3) $p \rightarrow q$
 - All three statements/propositions need to be true, hence p must be false (2.), which implies that q must be true (1.), consistent with a true implication in (3.)

Eks. fra opgave – systemspec.

- **"the message is scanned for viruses whenever the message was sent from an unknown system"**
 - p: "the message is scanned for viruses"
 - q: "the message was sent from an unknown system"
- How to phrase the intention mathematically? Consider:
 - 1) **If** "the message was sent from an unknown system" **then** "the message is scanned for viruses", i.e. $q \rightarrow p$ (if q, then p), versus
 - 2) "the message is scanned for viruses" **only if** "the message was sent from an unknown system", i.e. $p \rightarrow q$ (p only if q)

Øvelse: opskriv sandhedstabellen for de to betingede udsagn og identificér hvilke indgange der er problematiske i forhold til "intentionen" (eller konsekvensen) i sætningen ovenstående – tilsigtet/ok = sandt; utilsigtet/ikke ok = falskt.

Other implications / Definitions

”(You can receive an A in the course) only if (your score on the final is at least 90%)”: $p \rightarrow q$ (p only if q) if p you know that q, however when p is false, you may or may not have received a score at least 90%!

Conditional Statement	If p, then q	$p \rightarrow q$
Converse	If q, then p	$q \rightarrow p$
Inverse	If not p, then not q	$\sim p \rightarrow \sim q$
Contrapositive	If not q, then not p	$\sim q \rightarrow \sim p$
Biconditional	p if and only if q	$p \leftrightarrow q$

Source:

Bi-implication or biconditional statement

- Definition. Bi-implication $p \leftrightarrow q$ is a statement "p, if and only^{if} q".

- p if and only if q
- p is necessary and sufficient for q

$p \leftrightarrow q$ is exactly the same as
 $(\equiv) (p \rightarrow q) \wedge (p \leftarrow q)$

(and also the negation of the
Exclusive OR)

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

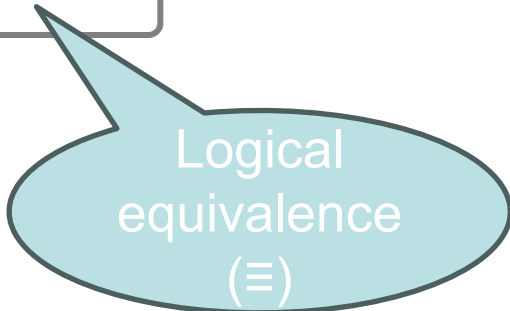
Ordbog

Bi-implication (engelsk) = Biimplikation (dansk)

Propositional equivalence

- Definition. Propositions p and q are equivalent, if they have the same truth tables.
- We can also give a definition in another way:
- p and q are equivalent, if $p \leftrightarrow q$ is a tautology

- $p \equiv q$



Logical
equivalence
($\equiv$)

Tautology and contradiction

- Tautology: a compound proposition that is always true
- Contradiction: a compound proposition that is always false
- Contingency: neither of

From previous slide:

" p and q are equivalent if $p \leftrightarrow q$ is a tautology"

Can we see this from the truth table of the biconditional?

De Morgan's laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

*De Morgan –
symbolic logic
developments*

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Distributive law

- Do you remember what the distributive law is about?

TABLE 5 A Demonstration That $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ Are Logically Equivalent.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Logical equivalences

Precedence of Logical Operators.	
Operator	Precedence
$\neg$	1
$\wedge$ $\vee$	2 3
$\rightarrow$ $\leftrightarrow$	4 5

$p \wedge \mathbf{T} \equiv p$ $p \vee \mathbf{F} \equiv p$	Identity laws
$p \vee \mathbf{T} \equiv \mathbf{T}$ $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv \mathbf{T}$ $p \wedge \neg p \equiv \mathbf{F}$	Negation laws

Example

Cond. statement/proposition

Law

$(p \wedge q) \rightarrow (p \vee q) \equiv \neg (p \wedge q) \vee (p \vee q)$ **(cond. disjunc. equiv.)**

$\equiv (\neg p \vee \neg q) \vee (p \vee q)$ De Morgan's (neg. conjunc.)

$\equiv \neg q \vee (\neg p \vee p) \vee q$ commutative and assoc.

$\equiv \neg q \vee \top \vee q$ negation

... thus a tautologi!

$\equiv (\neg q \vee q) \vee \top$ assoc.

$\equiv \top \vee \top$ negation

How to show equivalence?

- Approach 1. Using truth tables
- Approach 2. Developing a sequence of logical equivalences

Satisfiability

- A compound statement is satisfiable if there is an assignment to its variables that makes it true – either as a tautologi (truth values don't matter) or as a contingency (for some truth values)
 - Such an assignment is a solution
- ... unsatisfiable if the negation of the compound statement is a tautologi, i.e. all assignments of truth values to the variables makes the negation true
- Check by truth table (for low number of variables) or by logic reasoning (find one solution, show a contradiction)

*Lovelace – the
analytic engine*

Online learning

- Interactive demonstrations:
 - http://highered.mheducation.com/sites/125967651x/student_view0/interactive_demonstrations.html
- Self-assessment:
 - http://highered.mheducation.com/sites/125967651x/student_view0/self_assessments.html